



Shubham Kale

shubham23094@iiitd.ac.in

DOB: 01 April, 2000

Address: At post Rawande, Kopargaon,
Ahilyanagar District, Maharashtra

[K](#) Kaggle [G](#) Google-Scholar [G](#) Github [in](#) LinkedIn



Education

Indraprastha Institute of Information Technology, Delhi

M.Tech (Specialization in AI)

2023-2025

CGPA : 7.48

Dr.Babasaheb Ambedkar Technological University, Lonere

B.Tech, Information Technology

2018-2023

CGPA : 7.97

SSGM College, Kopargaon (Maharashtra)

Maharashtra State Board, Standard 12, PCM

2017-2018

Percentage : 81.85%

Shree Sharda English School, Kopargaon (Maharashtra)

Maharashtra State Board, Standard 10

2015-2016

Percentage : 89.20%

Skills

Expertise Area: Machine Learning, Deep Learning, Computer Vision, GenAI, NLP, LLM, MLOps

Programming Languages: Python, C++, C

Tools and Technologies: Kaggle, Pytorch, Streamlit, Hugging Face, Google Collab, Jupyter Notebook, Computer Vision Annotation Tool (CVAT), GAN, Transformers Models, Git, GitHub, AI tools, Visual Studio Code, Overleaf

Technical Electives: Machine Learning, Deep Learning, Statistical Machine Learning, Advance Machine Learning, Advances in Deep Learning.

Work Experience

Vehant Technologies Pvt Ltd Noida

Computer Vision Intern

(Feb, 25 – Sep, 25)

Worked on a **real-time person violence detection** system using **3D CNNs**, focusing on extracting **spatiotemporal features** from video. Developed and trained deep learning models for **violence classification**, optimized for **real-time inference** using **PyTorch** and **GPU acceleration**, and deployed in a **live surveillance pipeline**.

I also worked on a research project **state-of-the-art video classification model** for **multi-class video event detection and localization**, exploring advanced **temporal modeling techniques** to improve accuracy on benchmark video datasets.

(Sep, 25 – Present)

Tata Research Development and Design Centre - TCS Research

Researcher - DLAI

Working on **Large Reasoning Models (LRMs)** with a focus on alignment techniques such as **GRPO**, **DR-GRPO**, and **DAPO**, aiming to adapt model reasoning behavior toward **domain-specific AI4Research use cases**. Involved in designing training strategies to improve **reasoning accuracy**, **alignment**, and **generalization** for research-oriented tasks.

Publications

Advanced Smart City Monitoring: Real-Time Identification of Indian Citizen Attributes

Author : Shubham Kale, Shashank Sharma, Abhilash Khuntia

This project focuses on creating a smart surveillance system for Indian cities that can identify and analyze people's attributes in real-time. Using advanced technologies computer vision, and transformer architecture, the system can recognize attributes such as upper body color, what the person is wearing, accessories they are wearing, headgear, etc., and analyze behavior through cameras installed around the city. [PrePrint](#).

Author-3
(May, 24 - Jul, 24)

Real Time Emotion Analysis Using Deep Learning for Education, Entertainment, and Beyond

Author : Abhilash Khuntia, Shubham Kale

We developed a system that transforms facial expressions into emojis for instant feedback, enhancing interactions in online education, virtual events, and gaming. We utilized deep learning and real-time video processing to build a model for accurately categorizing facial expressions and integrated it into an app that analyzes live video to display corresponding emojis. [PrePrint](#).

Author-2
(Jan, 24 - Jul, 24)

UbiquitinAI: AI-Driven PTM Prediction in Protein Ubiquitination

Author : Shubham Kale, Prateek Paul, Dr. N.Arul Murugan

In this paper, we have focused on new ways for predicting ubiquitination sites in human proteins. Our approach was based on a dataset constructed previously, including protein sequences and identified ubiquitination sites. We considered different window sizes ranging from 7 to 99 amino acids to make accurate predictions of the ubiquitination site. We applied traditional machine learning and modern deep learning techniques to the problem. [Paper](#).

Author-3
(Jan, 24 - Jul, 24)

Projects

LLM-based Traffic Analysis System using CCTV Footage

Guide : Dr. Rajiv Ratan Shah (MIDAS Labs IIIT Delhi)

Developed an LLM-based traffic analysis system using CCTV footage for vehicle detection, accident detection, and real-time query responses. Utilized YOLOv5 for vehicle classification and QLoRA for model finetuning, improving accuracy by 20%. Built a multimodal pipeline for generating traffic reports and deployed the system as a chatbot for real-time user interaction. Successfully deployed the solution on Hugging Face, enhancing scalability for real-world traffic management applications.

Model Deployed: [Hugging Face](#)

Team Size-6
(Aug, 24 - Nov, 24)

NLP-based Vehicle Retrieval

Guide : Dr. AV Subramanyam (Asst. Prof & HOD ECE Dept at IIIT Delhi)

Developed an NLP-based vehicle retrieval system using Vision-Language Models (VLMs) such as CLIP, LLaMA, and Qwen2VL, leveraging contrastive learning techniques to align visual and textual features for improved retrieval accuracy. Curated a dataset of 10,000 images and captions from the Indian Driving Dataset (IDD) focusing on vehicle types like cars, buses, and trucks. Fine-tuned pre-trained models on this domain-specific dataset to achieve state-of-the-art performance in vehicle retrieval tasks specifically for Indian origin dataset.

Model Deployed: [Hugging Face](#)

Team Size-2
(Sep, 24 - Dec, 24)

Amazon ML Challenge - 2024

We developed a machine learning model to extract product details such as weight, dimensions, and voltage directly from images for e-commerce applications. Using PaddleOCR for text extraction and regex patterns to accurately identify and format values, we ensured precise predictions. We implemented parallel processing to efficiently handle large datasets, improving processing speed and scalability. This project showcased our ability to create effective, scalable AI solutions to address real-world challenges in the e-commerce industry.

Results : AIR 156

Team Size-2
(Sep, 24 - Sep, 24)

Course Undertaken

- Advance Machine Learning (Only MTech Student from Batch of 2025 to Complete the Subject)
- Advances in Deep Learning
- Large Language Model
- Bayesian Machine Learning
- Statistical Machine Learning
- Machine Learning
- Artificial Intelligence
- Capstone Project (ML/DL)
- Independent Project (NLP)
- Research Methodology

Positions of Responsibility

Overall Coordinator @ BACE DBATU Lonere	(Jun,22 – May,23)
Teaching (BSc CS/IT) @ Hirval Education Society Mahad	(May,23 - Jun,23)
Teaching Assistant (Communication Skills) @ IIIT Delhi	(Jun,23 - Dec,23)
Teaching Assistant (Natural Language Processing) @ IIIT Delhi	(Jan,24 - May,24)
Vehant Challenge 2024 SCSPAR-24 @ Team Lead	(May,24 - Jul,24)
Amazon ML Challenge 2024 @Team Lead	(Sep,24 - Sep,24)
Teaching Assistant (Machine Learning) @ IIIT Delhi	(Aug,24 - Dec,24)

Awards and Achievements

- GATE CSE 2022 94.73 Percentile
- GATE CSE 2023 89.62 Percentile
- Secured 2 Position in Vehant Challenge SCSPAR-24
- Received the Cash Price of 15000 for Second Position in SCSPAR-24 in IIST Thiruvananthapuram
- 2X Expert Position on Kaggle Platform
- Conference NCVPRIPG-24 IIST Thiruvananthapuram - Paper Presentation
- Amazon ML Challenge 2024 AIR 156 amongst 15000+ Team Registered
- OES ML Challenge Global Rank 4th

Interests and Hobbies

- Travelling
- Sports
- Spirituality

Declaration: The above information is correct to the best of my knowledge.

Shubham Kale

Date: February 3, 2026